Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories The goal is to find out more about spending habits, customer groups, product preferences, and subscription habits so that business decisions can be made more strategically.

2. Dataset Summary

- Rows: 3900

- Columns: 18

- Key Features:

  - Customer demographics (Age, Gender, Location, Subscription, Status)

  - Purchase Details (Item Purchased, Category, Purchase amount, Season, Size, Color)

  - Shopping Behavior (Discount Applied, promo code used, Previous Purchases, Frequency of purchases, Review Rating, Shipping type)

  - Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading**: Imported data using pandas.
- **Initial exploration**: used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data handling**: Checked for null values and imputed missing values in the Review Rating column using median rating for each product category.
- **Column Standardization**: Renamed columns for better readability and documentation.
- **Feature Engineering**:
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check**: Verified if discount_applied and promo_code_used were redundant, dropped promo_code_used.
- **Database Integration**: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis Using SQL

We performed structured analysis in PostgreSQL to answer key questions:

1. **Total Revenue by Gender**: Compared total revenue by male vs Female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High Spending Discount Users**: Identified customers who used discounts but still spend above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

Total rows: 839 Query complete 00:00:00.159

3. **Top 5 products by Review Rating**: Found products with the highest average review rating.

	item_purchased text	Average Product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping type comparison**: Compared average purchase amounts between standard and express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs Non- Subscribers**: Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount dependent Products**: Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation**: Classified customers into New, Returning and Loyal segments based on the pervious history.

	customer_segment text	Number of Customers bigint
1	Royal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per category**: List of the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers and subscriptions**: Checked whether customers with more than 5 purchases are more likely to subscribe.

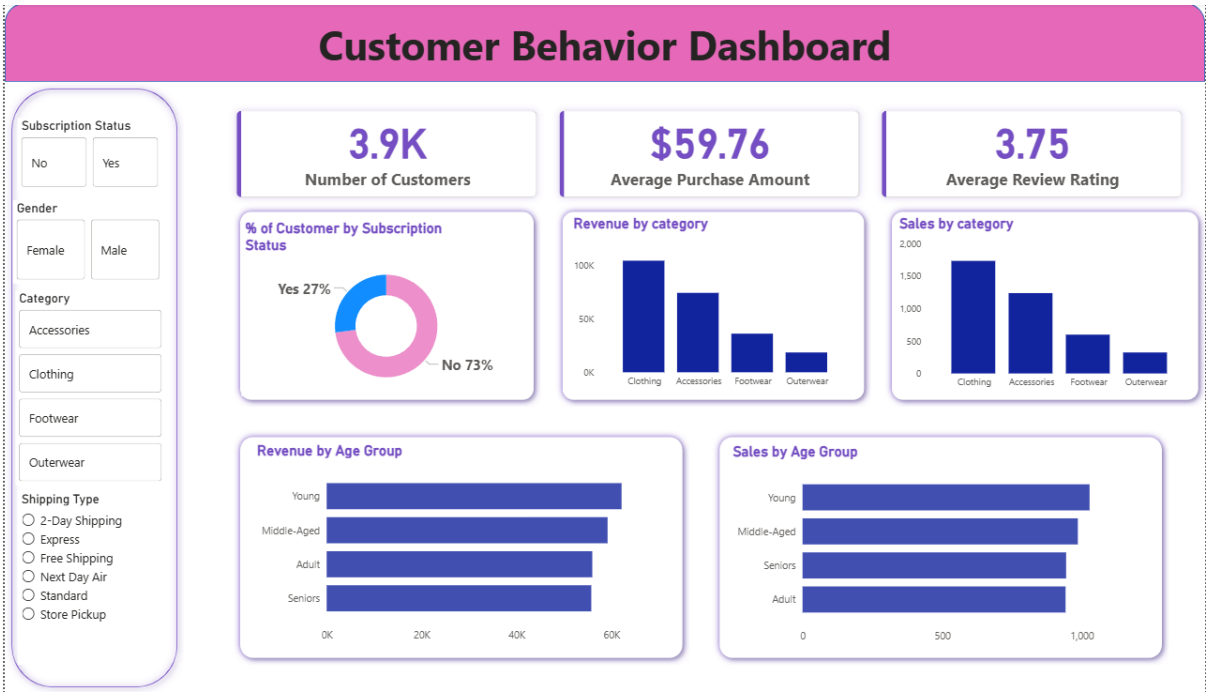
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age group**: Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young	62143
2	Middle-Aged	59197
3	Adult	55978
4	Seniors	55763

5. Dashboard in Power BI

Finally, built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Get More Subscribers:** Offer special benefits to subscribers to get more of them.
- **Programs to keep customers coming back:** Give rewards to customers who buy from you again to move them into the "Loyal" group.
- **Review Discount Policy**: Balance sales boots with margin control.
- **Product Positioning**: In campaigns, make sure to show off the best-selling and highest-rated products.
- **Targeted marketing**: Put more effort into high-revenue age groups and people who use express shipping.